SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING

CSA47 08– DEEP LEARNING

ASSIGNMENT – 1

CO1 – MATHEMATICAL FOUNDATIONS AND MACHINE LEARNING PRINCIPLES

Scenario: Healthcare Risk Prediction

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A hospital is building an intelligent system to predict disease risk from age, blood pressure, glucose level, cholesterol and family history. This report answers Tasks 1 to 12 using definitions, formulae, step-by-step calculations and healthcare interpretation.

TASK 1 – Patient Data Representation and Normalization (5 Marks)

(a) Effect of using raw features. In the sample below, age lies between 25 and 60 years, blood pressure (BP) between 120 and 160 mmHg, glucose between 90 and 180 mg/dL and cholesterol between 180 and 260 mg/dL. Because the features have different units and ranges, using raw values causes the following problems:

Table 1: Raw patient data

Patient	Age (years)	BP (mmHg)	Glucose (mg/dL)	Cholesterol (mg/dL)
Patient 1	25	120	90	180
Patient 2	60	130	180	200
Patient 3	45	160	130	260

- **Large-scale features dominate.** Dot products, distances and gradients are driven by the feature with the largest numeric spread. The raw Euclidean distance between Patient 1 and Patient 2 is $\sqrt{(35^2 + 10^2 + 90^2 + 20^2)} = \sqrt{9825} \approx 99.12$, and glucose alone contributes 8100/9825 ≈ 82% of the squared distance, although age and blood pressure are clinically just as important.
- **Unit dependence.** Reporting glucose in mmol/L instead of mg/dL would change the model output for the same patient.
- **Slow, unstable optimization.** Uncentred features with very different variances give an elongated (ill-conditioned) loss surface, so gradient descent zig-zags and needs a very small learning rate. Large inputs can also saturate sigmoid or tanh units.
- **Non-zero means.** Uncentred features become strongly correlated with the bias term, which further slows learning.

(b) Feature matrix and normalization. Let x_{ij} be the value of feature j for patient i. The records form the feature matrix $X \in R^{3\times4}$, with rows as patients and columns as (Age, BP, Glucose, Cholesterol):

$$X = [\ 25\ 120\ 90\ 180\ ;\ \ 60\ 130\ 180\ 200\ ;\ \ 45\ 160\ 130\ 260\ ]$$

Min–max normalization rescales every column to the range [0, 1]:

$$x'_{ij} = (x_{ij} - \min_j) / (\max_j - \min_j)$$

- **Step 1 – Column minima and maxima:** Age 25 and 60; BP 120 and 160; Glucose 90 and 180; Cholesterol 180 and 260.

- **Step 2 – Apply the formula:** for example, Patient 3 age = (45 − 25)/(60 − 25) = 20/35 = 0.571; Patient 2 BP = (130 − 120)/(160 − 120) = 10/40 = 0.250; Patient 3 glucose = (130 − 90)/90 = 0.444; Patient 2 cholesterol = (200 − 180)/80 = 0.250.
- **Step 3 – Normalized matrix** (Table 2).

Table 2: Normalized feature matrix X′

Patient	Age	BP	Glucose	Cholesterol
Patient 1	0.000	0.000	0.000	0.000
Patient 2	1.000	0.250	1.000	0.250
Patient 3	0.571	1.000	0.444	1.000

Numerical verification. Every column of X′ has minimum 0 and maximum 1, so all features are on the same scale. As an alternative, centring and standardization $z = (x - \mu)/\sigma$ gives, for age, $\mu = 43.33$ and $\sigma = 14.34$, so $z = -1.279$, 1.162 and 0.116 for the three patients; these sum to approximately 0, confirming the column is centred. After min–max scaling, the distance between Patient 1 and Patient 2 becomes $\sqrt{(1^2 + 0.25^2 + 1^2 + 0.25^2)} = \sqrt{2.125} \approx 1.458$, and no feature dominates merely because of its units.

Interpretation. Patient 1 has the lowest value on every feature, Patient 2 has the highest age and glucose, and Patient 3 has the highest blood pressure and cholesterol. The normalized values show each patient's relative position within the observed range of each feature.

(c) Importance for learning methods. Distance-based methods (k-nearest neighbours, K-means, RBF-kernel SVM) use $d(x, y) = \sqrt{\Sigma(x_j - y_j)^2}$. Without normalization, the feature with the largest scale decides which patients are "near". Normalization gives every feature comparable influence so that similarity reflects the whole patient profile. **Gradient-based methods** (logistic regression, neural networks) update each weight using $\partial L/\partial w_j = (\hat{y} - y) \cdot x_j$, so the gradient size depends directly on the feature scale. Centred, equally scaled features give a rounder loss surface, which permits a larger learning rate, faster and more stable convergence, and weights of comparable size, so that L2 regularization treats all features fairly.

TASK 2 – Vector Representation and Matrix Operations (2 Marks)

(a) Vector formulation. Take the feature order (Age, BP, Glucose, Cholesterol). A patient is the column vector $x = [x_1, x_2, x_3, x_4]^T$ and the importance weights are $w = [w_1, w_2, w_3, w_4]^T$, with $w_j \geq 0$ and $\Sigma w_j = 1$. The preliminary risk score is the dot product:

$$s = w^T x = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

For n patients stored in the feature matrix X (n × 4), all scores are computed together as the matrix–vector product **s = Xw**, where s is an n × 1 vector.

(b) Numerical example. Use the normalized values from Table 2 and give glucose the highest importance: $w = [0.2, 0.2, 0.4, 0.2]^T$. For Patient 3, $x = [0.57, 1.00, 0.44, 1.00]^T$:

$$s = 0.2(0.57) + 0.2(1.00) + 0.4(0.44) + 0.2(1.00) = 0.114 + 0.200 + 0.176 + 0.200 = 0.690$$

For all three patients, $s = Xw$ gives $s_1 = 0.2(0) + 0.2(0) + 0.4(0) + 0.2(0) = 0.000$ and $s_2 = 0.2(1) + 0.2(0.25) + 0.4(1) + 0.2(0.25) = 0.700$, so $s = [0.000, 0.700, 0.690]^T$.

Interpretation. Because the weights sum to 1, the score lies between 0 and 1, and a higher score indicates higher preliminary risk. Patients 2 (0.700) and 3 (0.690) have almost equal scores and are far above Patient 1 (0.000), so they would be prioritized for further examination. The score is a simple linear indicator, which is exactly the computation performed by a single neuron before its activation function; the weights could later be learned from labelled data.

TASK 3 – Probability and Conditional Probability (6 Marks)

Events. Let D = the patient has the disease and E = the patient has elevated glucose. Total patients N = 1000.

Table 3: Contingency table

Group	Elevated glucose (E)	Not elevated (E^{c})	Total
Disease (D)	160	40	200
No disease (D^c)	120	680	800
Total	280	720	1000

(a) P(E). Using the total number of patients with elevated glucose: $P(E) = (160 + 120)/1000 = 280/1000 =$ **0.28 (28%)**. Check by the law of total probability: $P(E) = P(E|D)P(D) + P(E|D^c)P(D^c) = 0.80 \times 0.20 + 0.15 \times 0.80 = 0.16 + 0.12 = 0.28$.

(b) P(D|E). Using $P(D|E) = P(D \cap E)/P(E) = 0.16/0.28 = 160/280 \approx$ **0.571 (57.1%)**.

(c) P(E|D). Using $P(E|D) = P(D \cap E)/P(D) = 160/200 =$ **0.80 (80%)**.

Note that $P(D|E) \neq P(E|D)$. The first answers "given high glucose, how likely is disease?" (57.1%); the second answers "given disease, how likely is high glucose?" (80%).

(d) Healthcare interpretation. The overall disease rate is 20%, but among patients with elevated glucose it rises to 57.1%, about 2.9 times higher. Among patients without elevated glucose it is only $P(D|E^c) = 40/720 \approx 5.6\%$. Conditional probabilities therefore rank patients by risk: those with elevated glucose are referred first for confirmatory tests, which is useful when testing resources are limited. Since $P(E|D) = 0.80$, 20% of diseased patients have normal glucose, so glucose alone cannot rule disease out and further features should be combined.

TASK 4 – Bayesian Statistics for Disease Classification (4 Marks)

Given. Let D = disease and + = positive test. Prior $P(D) = 0.10$ and $P(D^c) = 0.90$. Likelihoods: $P(+|D) = 0.90$ and $P(+|D^c) = 0.05$.

(a) Posterior probability. Bayes' theorem states:

$$P(D|+) = P(+|D)\cdot P(D) / P(+), \quad \text{where} \quad P(+) = P(+|D)P(D) + P(+|D^c)P(D^c)$$

- Step 1: $P(+) = 0.90 \times 0.10 + 0.05 \times 0.90 = 0.090 + 0.045 = 0.135$.
- Step 2: $P(D|+) = 0.090/0.135 = 0.667$.

The probability that a patient has the disease given a positive test is **66.7%**.

(b) Roles of prior and likelihood. The **prior** P(D) is the belief about disease before seeing the test, here the 10% prevalence. The **likelihood** P(+|D) measures how strongly the observed result is expected if the patient is diseased compared with $P(+|D^c)$ if not. The posterior is proportional to likelihood × prior. The likelihood ratio is 0.90/0.05 = 18 and the prior odds are 0.10/0.90 = 1/9, so the posterior odds are 18/9 = 2, which equals a probability of 2/3. Although the test is accurate, the posterior is only 66.7% because the disease is rare: out of 100 people, 9 true positives are accompanied by 4.5 false positives from the large healthy group.

(c) Usefulness under uncertainty. Bayesian reasoning combines prior medical knowledge (prevalence, earlier findings) with new evidence in a principled way. It outputs probabilities rather than hard labels, so decisions can use risk thresholds; it avoids the base-rate fallacy; it updates sequentially, since the posterior after one test becomes the prior for the next; and it works with small or noisy data. It is the basis of classifiers such as Naive Bayes, which are widely used for medical diagnosis.

TASK 5 – Entropy and Information Gain (3 Marks)

(a) Entropy. For a classification with class proportions p_i, the entropy is $H(S) = -\Sigma\, p_i \log_2 p_i$. Here p(high-risk) = 40/100 = 0.4 and p(low-risk) = 0.6:

$$H(S) = -0.4 \log_2(0.4) - 0.6 \log_2(0.6) = 0.4(1.3219) + 0.6(0.7370) = 0.5288 + 0.4422 \approx 0.971 \text{ bits}$$

(b) Interpretation. For two classes entropy ranges from 0 (all records in one class, no uncertainty) to 1 bit (50:50, maximum uncertainty). A value of 0.971 is close to the maximum, so the class of a randomly chosen record is hard to predict from the class proportions alone; on average almost one yes/no question is needed to determine the risk class.

(c) Information gain. Information gain measures the expected reduction in entropy after splitting the data on a feature A: $IG(S, A) = H(S) - \Sigma_v\, (|S_v|/|S|)\, H(S_v)$. As a hypothetical illustration, suppose splitting on glucose gives the groups in Table 4.

Table 4: Illustrative glucose split

Group	Records	High-risk	Low-risk	Entropy
High glucose	50	35	15	0.881
Normal glucose	50	5	45	0.469

The weighted entropy after the split is 0.5(0.881) + 0.5(0.469) = 0.675, so IG = 0.971 − 0.675 ≈ **0.296 bits**. The feature with the highest information gain removes the most uncertainty and is therefore chosen as the best split in a decision tree or ranked highest during feature selection. A large gain for glucose would show that it is a useful predictor of risk.

TASK 6 – Numerical Computation and Distance Measurement (3 Marks)

(a) Euclidean distance. For normalized vectors A = (0.70, 0.80, 0.60) and B = (0.50, 0.75, 0.90) in the order (glucose, BP, cholesterol), $d(A, B) = \sqrt{\Sigma(a_j - b_j)^2}$.

- Differences: 0.70 − 0.50 = 0.20; 0.80 − 0.75 = 0.05; 0.60 − 0.90 = −0.30.
- Squares: 0.0400; 0.0025; 0.0900; sum = 0.1325.
- $d(A, B) = \sqrt{0.1325} \approx$ **0.364**.

(b) Similarity. A small distance means the patient records are close in feature space and therefore similar. The largest possible distance in the unit cube is $\sqrt{3} \approx 1.732$, so 0.364 (about 21% of the maximum) indicates fairly similar patients; the main difference is cholesterol, which contributes 0.09/0.1325 ≈ 68% of the squared distance. Such distances allow k-nearest neighbours to predict a patient's risk from similar patients, and allow clustering algorithms to group similar records.

(c) Need for normalization. Euclidean distance adds squared differences, so a feature measured on a large scale produces large differences and dominates the sum, exactly as glucose did in Task 1 (82% of the squared distance). Normalization places glucose, blood pressure and cholesterol on a common scale so that each contributes according to its relative difference, and the distance is independent of the measurement units.

TASK 7 – Bias–Variance Trade-off (4 Marks)

(a) Identification.

Table 5: Diagnosis of the three models

Model	Training error	Validation error	Diagnosis
Model A	High	High	High bias (underfitting)
Model B	Very low	High	High variance (overfitting); large gap
Model C	Low	Low, similar	Good balance of bias and variance

(b) Effect of model complexity. The expected prediction error decomposes as: Expected error = Bias² + Variance + Irreducible noise. A simple model makes strong assumptions, so it has high bias (it misses the true pattern) but low variance (its predictions change little with different training sets). As complexity increases, bias falls because the model can represent more complicated relationships, but variance rises because the model starts to fit the noise of the particular training sample. The total error is therefore U-shaped, and the best model lies at intermediate complexity, as for Model C.

(c) Improvements.

- **Model A (high bias):** increase capacity (more layers or neurons, polynomial or interaction features), add informative features such as family history, reduce the regularization strength and train longer.

- **Model B (high variance):** collect more training data, apply L1/L2 regularization or dropout, simplify the model, use early stopping, remove noisy features, use cross-validation and ensemble methods such as bagging.

TASK 8 – Overfitting, Underfitting and Model Capacity (4 Marks)

(a) Identification.

Table 6: Model behaviour

Model	Behaviour
Simple model	Underfitting: cannot capture disease patterns; poor on both training and unseen data
Moderately complex model	Appropriate generalization: consistent performance on training and unseen patients
Highly complex model	Overfitting: near-perfect training accuracy but poor on unseen patients

(b) Effect of capacity. Model capacity is the range of functions a model can represent, and it grows with the number of parameters, layers or polynomial degree. A low-capacity model, such as a straight-line classifier, cannot represent non-linear relationships like the interaction between age and glucose. A higher-capacity model can represent such complex relationships and therefore learn the disease pattern more accurately.

(c) Capacity and generalization. At first, additional capacity removes the model's inability to represent the true pattern, so both training and validation errors fall. Beyond the optimal capacity, the extra flexibility is used to memorize noise and individual quirks of the training patients. Training error keeps decreasing, but validation error rises, so the generalization gap (validation error − training error) widens, giving the characteristic U-shaped validation curve. Capacity should therefore be chosen using validation data, and controlled using regularization, early stopping and more training data.

TASK 9 – Supervised Learning and Model Evaluation (3 Marks)

(a) Why supervised. Each historical record has input features x and a known label y (diseased or disease-free). The system learns a mapping f: x → y from these labelled input–output pairs and uses it to predict labels for new patients. Predicting a discrete category makes this a supervised binary classification problem.

(b) Training and testing sets. The **training set** (typically 70–80% of records) is used to learn the model parameters. The **testing set** contains records the model has never seen and is used once, at the end, to estimate how well the model generalizes; evaluating on training data would be over-optimistic because the model may have memorized it. A separate validation set (or cross-validation) is used to tune hyperparameters. The split should be stratified to preserve the disease proportion, and normalization statistics should be computed only from the training set to avoid data leakage.

(c) Accuracy. Accuracy = number of correct predictions / total predictions = 92/100 = **0.92 (92%)**. The model classifies 92 of 100 unseen patients correctly and makes 8 errors. Accuracy alone can mislead when classes are imbalanced: if only 8% of patients are diseased, a model that always predicts "disease-free" also scores 92%. Because a missed diseased patient (false negative) is costly, the confusion matrix, sensitivity (recall), specificity, precision and F1-score should also be reported.

TASK 10 – Unsupervised Learning and Patient Grouping (4 Marks)

(a) Why unsupervised. The records contain only features (age, blood pressure, glucose, cholesterol) and no risk labels, so there is no target to predict. Unsupervised learning discovers hidden structure in unlabelled data, and clustering can reveal natural patient groups such as low-risk, borderline and high-metabolic-risk profiles without requiring costly labelling.

(b) K-means procedure. K-means partitions the patients into K clusters:

- **Step 1:** Normalize all features so that each has comparable scale.
- **Step 2:** Choose K (for example using the elbow method or the silhouette score).
- **Step 3:** Initialize K centroids (randomly or with k-means++).
- **Step 4 – Assignment:** assign each patient to the nearest centroid using Euclidean distance.
- **Step 5 – Update:** recompute each centroid as the mean of its members, $\mu_k = (1/|C_k|) \Sigma_{x \in C_\{k\}} x$.
- **Step 6:** Repeat Steps 4 and 5 until the assignments stop changing. This minimizes $J = \Sigma_k \Sigma_{x \in C_\{k\}} ||x - \mu_k||^2$, although only a local minimum is guaranteed, so several initializations are run.

For example, with normalized centroids $c_1 = (0.2, 0.3, 0.2, 0.2)$ and $c_2 = (0.8, 0.7, 0.9, 0.8)$, a patient $x = (0.7, 0.6, 0.8, 0.7)$ has $d(x, c_1) = \sqrt{(0.25 + 0.09 + 0.36 + 0.25)} = \sqrt{0.95} \approx 0.975$ and $d(x, c_2) = \sqrt{(4 \times 0.01)} = 0.200$, so the patient is assigned to cluster 2.

(c) Distance as similarity. The Euclidean distance $d(x, y) = \sqrt{\Sigma(x_j - y_j)^2}$ measures how different two patients are across all features: a small distance means similar characteristics. K-means places patients with small distances to the same centroid, giving clusters with small within-cluster distance and large between-cluster distance. Normalization is needed so that no feature dominates this distance.

(d) Clinical use. The mean feature values of each cluster can be examined; for example, a cluster with high average glucose and cholesterol may be flagged for earlier screening, lifestyle advice or closer monitoring, and unusual small clusters may point to rare conditions or data errors. The clusters are only hypotheses about patient similarity: they carry no diagnostic label, so they must be reviewed and validated by clinicians and confirmed with clinical tests. They support, but do not replace, medical diagnosis, and the final decision for each patient remains with the doctor.

TASK 11 – Comparison of Supervised and Unsupervised Learning (4 Marks)

(a) Suitable paradigm. Dataset A (features with disease labels) suits **supervised learning** because the labels provide the correct answer for each record, so a classifier can learn to map features to disease status. **Dataset B** (features without labels) suits **unsupervised learning**, such as clustering or dimensionality reduction, because there is no target and the aim is to discover structure.

(b) Comparison.

Table 7: Supervised versus unsupervised learning

Aspect	Supervised (Dataset A)	Unsupervised (Dataset B)
Data	Features X and labels y	Features X only
Objective	Learn f: X → y that predicts the label accurately	Discover groups, patterns or low-dimensional structure
Output	Predicted class or probability of disease	Cluster assignments, centroids, reduced features
Examples	Logistic regression, decision tree, neural network	K-means, hierarchical clustering, PCA
Evaluation	Accuracy, recall, precision, F1 on test data	Silhouette score, within-cluster distance, expert review

(c) Recommendation. To predict disease status for a new patient, a **supervised classifier trained on Dataset A** is recommended, for example regularized logistic regression or a decision tree with normalized inputs, with its capacity chosen by cross-validation. Only labels provide the error signal needed to predict a specific outcome, and the model can return the probability P(disease | x) for prioritizing patients. Clustering alone cannot output "diseased" without labels, and clusters need not coincide with disease status. Dataset B can still help as a supporting resource, for example by creating cluster-based features or by semi-supervised learning, but the prediction itself should come from the supervised model.

TASK 12 – Design Challenge: Selecting and Adapting a Learning Strategy (8 Marks)

A. Impact Analysis (2 Marks)

(a) Effect of dataset size. Growing from 10,000 to 1,000,000 records multiplies the number of rows of the feature matrix by 100.

- **Data representation:** with five double-precision features the matrix needs about $10^6 \times 5 \times 8$ bytes = 40 MB, which fits in memory, but additional features or indicator columns can exceed it. Single precision, chunked loading and sparse formats become important.
- **Numerical computation:** matrix–vector products and K-means iterations scale linearly with n (O(nd) and O(nKd) per pass), but pairwise distances for k-nearest neighbours scale as $O(n^2d)$, about 10^{12}

operations for one million records, which is impractical without approximate neighbour search. Summing millions of terms also accumulates floating-point rounding error, so numerically stable algorithms are needed.

- **Learning efficiency:** full-batch gradient descent becomes slow, so mini-batch SGD and mini-batch K-means are used. More data lowers variance, so a model of higher capacity can be supported without overfitting.

(b) Normalization, missing values and variability. Normalization keeps features on comparable scales for distance and gradient computations; its statistics (minimum, maximum, mean, standard deviation) must be computed on training data only, and robust scaling or clipping is needed because outliers become more likely in large data. **Missing values** are certain in one million records; deleting incomplete rows wastes data and can bias results, since missingness may be related to the patient's condition, so imputation (mean, median or model-based) plus a missing-value indicator is preferred. **Data variability** from different hospitals, laboratories, age groups and measurement errors increases variance and noise, so robust preprocessing and evaluation across subgroups are essential.

B. Evaluation of the Original Approach (2 Marks)

(a) Is a single supervised approach sufficient? No. A supervised model predicts a known target (disease risk) from labelled examples, but patient-group discovery has no predefined target; it requires finding unknown structure with unsupervised methods such as clustering. A classifier's output (a class or a probability) does not reveal which patients share similar profiles, so one supervised approach cannot meet both requirements.

(b) Limitations of labelled learning. Labels require confirmed diagnoses and follow-up, so only a small fraction of 1,000,000 records is likely to be labelled. Ignoring unlabelled records discards information about the true distribution of the features; labelled records may be a biased sample; labels may be noisy; and a model trained only on predefined labels cannot discover new patient subtypes.

C. Adaptation or Redesign (2 Marks)

(a) Proposed strategy. The analytical strategy should be **modified**: the supervised classifier is retained, and an unsupervised stage plus scalable preprocessing is added.

- **Step 1 – Preprocessing:** clean the data, impute missing values, add missing indicators and normalize using training statistics.
- **Step 2 – Unsupervised stage:** run mini-batch K-means on all records (labelled and unlabelled) to discover patient groups; select K with the elbow method and silhouette score.
- **Step 3 – Feature augmentation:** add the cluster membership and the distances to the centroids as extra features.

- **Step 4 – Supervised stage:** train a regularized classifier (for example logistic regression or a small neural network) on the labelled records to output P(disease | x). Confident predictions on unlabelled records can optionally be used as pseudo-labels (semi-supervised learning).
- **Step 5 – Use:** rank patients by predicted risk for testing, and use clusters for screening programmes, with clinicians reviewing all decisions.

(b) Incorporating the mathematical components. Probability: the classifier outputs a probability, and Bayes' theorem with the disease prevalence as prior is used to interpret it and to choose a decision threshold that favours sensitivity. **Distance measures:** Euclidean distance drives K-means and cluster assignment. **Normalization:** puts all features on a common scale so that the distances and gradients are balanced. **Model capacity:** chosen by validation, with regularization; because the larger data supports more complexity, capacity can be increased gradually while the validation error is monitored.

D. Evidence-Based Defence (2 Marks)

(a) Experiments and evaluation measures. The designs are compared on identical splits using stratified 5-fold cross-validation and a held-out test set.

Table 8: Experimental plan

No.	Experiment	Purpose and measures
E1	Original: supervised classifier on labelled records only	Baseline accuracy, recall, precision, F1, AUC-ROC
E2	Redesign: preprocessing + clusters as features + classifier	Test whether clustering improves prediction and calibration (log-loss)
E3	Redesign + pseudo-labelling of unlabelled records	Measure the benefit of unlabelled data
E4	Scalability: run E1–E3 on 10,000; 100,000; 1,000,000 records	Training time, memory, learning curves
E5	Clustering quality of the K-means stage	Silhouette score, within-cluster distance, stability across runs, clinical review

Results are reported as mean ± standard deviation across folds, and paired tests are used to judge whether differences between designs are significant. Sensitivity and recall are emphasized because missed cases are costly.

(b) Justification of the final design. Mathematical reasoning: the bias–variance trade-off shows that more data reduces variance, so a regularized moderate-capacity model generalizes well; Bayes' theorem ensures predictions respect the low base rate; each K-means iteration cannot increase the objective J, so it converges, and its cost O(nKdt) is linear in n, unlike the $O(n^2)$ cost of pairwise methods. **Generalization:** performance

is judged on unseen test patients, and the gap between training and validation error is monitored. **Computational needs:** mini-batch SGD and mini-batch K-means keep training feasible for one million records. **Data characteristics:** missing values, mixed scales and a mixture of labelled and unlabelled records are handled by imputation, normalization and the combined supervised–unsupervised design. The final decision is therefore to **modify** the original strategy into a hybrid pipeline: supervised classification for disease-risk prediction and unsupervised clustering for patient-group discovery.